

Expected Questions from Rotational Motion Discussion

1. Two bodies of mass M & m [$M > m$] are joined by a string, then center of mass lies
 - a. Towards M
 - b. Between M & m
 - c. Towards m
 - d. Both 'a' and 'b'
2. The center of mass is independent of
 - a. Position vector
 - b. Relative separation of distance
 - c. Masses of particle
 - d. Force of the particle
3. If a gymnast on a rotating stool with his arms outstretched suddenly lower his arms:
 - a. The angular velocity decreases
 - b. The moment of inertia decreases
 - c. The angular velocity remains constant
 - d. The angular momentum increases
4. If no internal force is applied in a body the velocity of the center of mass is :
 - a. Zero
 - b. Increases
 - c. Decreases
 - d. Remains constant
5. The product of moment of inertia & angular acceleration gives
 - a. Linear momentum
 - b. Angular momentum
 - c. Torque
 - d. Force
6. Angular momentum of a body is defined as the product of:
 - a. Mass and angular velocity
 - b. Linear velocity and angular velocity
 - c. Centripetal force and radius
 - d. Moment of inertia and angular velocity
7. The moment of momentum is called
 - a. Couple
 - b. Liquid
 - c. Impulse
 - d. Angular momentum
8. A ring, a disc, solid sphere, hollow sphere are dropped from the same inclined plane of same height then which one of the following reaches the ground first
 - a. Ring
 - b. Disc
 - c. Solid sphere
 - d. Hollow sphere
9. The moment of inertia of a body of mass M about a given axis is I . What is the radius of gyration?
 - a. $1/M$
 - b. $1M$
 - c. $\sqrt{I/M}$
 - d. $\sqrt{IM}$
10. The torque due to gravitational force on a body about its center of mass is
 - a. Infinite
 - b. Finite
 - c. Zero
 - d. Cannot be measured
11. Two-point masses of 1 kg and 2 kg separated by 0.5m constitute a system. The distance of the center of mass of system from 1kg mass is
 - a. 0.15 cm
 - b. 0.25 cm
 - c. 0.33 cm
 - d. 0.4 cm
12. An inclined plane makes an angle of 30° with horizontal. A solid cylinder rolling down this inclined plane from rest without slipping has linear acceleration equal to
 - a. $\frac{g}{3}$
 - b. $\frac{g}{2}$
 - c. $\frac{2g}{3}$
 - d. $\frac{g}{4}$
13. A rest projectile suddenly explodes into two fragments then the direction of center of mass is
 - a. Parabolic
 - b. St. line
 - c. Hyperbolic
 - d. Rest

14. A body is moving with moment of inertia is 400kg m^2 is rotating with angular velocity 20 rad/s . Calculate the torque of the body
- a. 2000 N/m
 - b. 4000 N/m
 - c. 8000 N/m
 - d. 16000 N/m
15. What happens to moment of inertia when polar ice melts?
- a. Increases
 - b. Decreases
 - c. Remains constant
 - d. None of these
16. Ice skaters use the principle of _____ for spinning.
- a. Moment of inertia
 - b. Angular momentum
 - c. Torque
 - d. Radius of gyration
17. A cylinder is rotated in inclined plane, find its acceleration.
- a. $g\sin\theta$
 - b. $\frac{2}{3}g\sin\theta$
 - c. $\frac{1}{2}g\sin\theta$
 - d. $\frac{3}{2}g\sin\theta$
18. A person standing on a rotating platform has his hands out stretches suddenly lowers his arms then angular momentum of platform:
- a. Increases
 - b. Decreases
 - c. Remains constant
 - d. Becomes zero
19. The spokes are used in bicycle wheel to
- a. Increase frictional force
 - b. Decrease frictional force
 - c. Increase moment of inertia
 - d. Increase angular momentum
20. Three identical rods each of mass 'm' and length 'l' are joined to form a triangle. Moment of inertia about its base is
- a. $\frac{ml^2}{3}$
 - b. $\frac{2ml^2}{3}$
 - c. $\frac{3ml^2}{4}$
 - d. ml^2